BPDRPT AZD9592 Conjugation Process Description

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1 INTRODUCTION

1.1 Purpose

This report describes the conjugation process and process parameters for the production of AZD9592 Drug Substance lot. The overall process is designated “Process 1”.

Table 1.1 General Properties: AZD9592 Drug Substance

Property	Description
Product classification	Antibody Drug Conjugate
Payload	SG3932
Target DAR	6.0
Isoelectric point (pI) range	8.1 – 9.0
Extinction coefficient mAb Conjugated ADC	mAb: 1.51 $(mg/ml)^{-1}cm^{-1}$ ADC: 1.81$(mg/ml)^{-1}cm^{-1}$
mAb Intermediate molecular weight (g/mol)	144,483.9 Da (144,482.42 theoretical)
Payload molecular weight (g/mol)	1148.3
mAb Intermediate concentration (g/L)	50 (range: 45-55)
Overall expected conjugation yield (%)	85%
Formulated Drug Substance concentration (g/L)	20.0
Drug Substance formulation	20 mM histidine/histidine-HCl, 240 mM sucrose, 0.02% (w/v) polysorbate 80, pH 6.0
Drug substance presentation	Stored frozen at ≤ - 80°C in HDPE bottles in light protecting bag.

1.2 Assumptions

- This report does not consider scale or facility constraints.
- Unless otherwise noted, procedures pre/post common unit operations should be followed for filters and unit operations per DEV000_GB_0469.
- Unless otherwise noted, all conductivity values are compensated to a reference temperature of 25°C using a temperature coefficient (θ) of 2.0% per °C.
- The target volumes are expected to be sufficient. Overage to account for system hold up volume is site dependent and should be a separate phase and not added into the volume targets for the process.
- Unless otherwise noted, flow rate ranges are for steady state operation and do not consider pump ramping strategy.
- Unless otherwise noted, all unit operations are conducted at controlled room temperature (15-27°C).
- For facilities for which 1.0 N NaOH is not feasible, 0.5 N NaOH may be used as a substitute with “Conductivity (mS/cm @ 25°C), final” NOR: 84 – 113.
- Unless otherwise noted, all chromatography step phases are operated in down flow mode.
- Unless otherwise noted, all process parameters are inputs.
- Safety practices and procedures defined by SHE should be followed at all times and can supersede details described in this document.

1.3 Abbreviations/Acronyms/Definition

Table 1.3-1 Abbreviations and Acronyms

Acronym	Definition	Acronym	Definition
AEX	Anion exchange chromatography	LMH	Liters/meter^2-hour
ADC	Antibody Drug Conjugate	MV	Membrane volume
AL	Alert limit	NA	Not applicable
Bldg	Building	NC	Non-conformance
BR	Batch record	NOR	Normal operating range
CEX	Cation exchange chromatography	OD	Optical density
Conc.	Concentration	PCR	Polymerase chain reaction
CV	Column volume	PES	Polyethersulfone
DAR	Drug Antibody Ratio	PFB	Pre-formulated bulk
DBC	Dynamic binding capacity	P/N	Part number
DS	Drug substance	PP	Polypropylene
DV	Diafiltration volumes	psi; psig	pounds per square inch; pounds per square inch gauge
EAB	Excipient addition buffer	qs	Bring to final weight (quantum satis)
ELISA	Enzyme-linked immunosorbent assay	S.F.	Safety factor
FIO	For information only	SS	Stainless steel
HDPE	High density polyethylene	TBD	To be determined
HETP	Height equivalent to a theoretical plate	TMP	Trans-membrane pressure
HPLC	High-performance liquid chromatography	UDS	Unformulated drug substance
HPSEC	High-performance size-exclusion chromatography	UF/DF	Ultrafiltration/diafiltration
HCP	Host cell protein	WFI	Water for injection
IEC	Ion exchange chromatography		
LG	Linear gradient		
LR	Lot release		

1.4 Definitions of Terminologies for Parameter Ranges

1.4.1 Target

The target value is a preset value that a process parameter is aimed at by process controls. The controls include any combination of manual or automated, discrete or continuous, feed-forward or feed-back responses to operate the process near the target value. Where Targets are provided, the process should be controlled to meet the target value as closely as possible, not just controlling to the edge of the Alert Limit or Normal Operating Range.

1.4.2 Alert Limit (AL)

This is the expected process range. Exceeding this limit suggests that corrective action is required to prevent the process from exceeding the normal operating range (NOR) or other process constraints. Sending Unit and subject matter experts should be informed to determine if any corrective action is required. Exceeding this limit does not trigger a non-conformance investigation.

1.4.3 Normal Operating Range (NOR)

Operating parameter ranges, also called Action Limits, are established based upon current understanding and process knowledge gained through development and process characterization. NORs are specified in manufacturing batch records and/or other GMP documentation (such as automation recipes) to control the process. A value exceeding a NOR triggers a non-conformance investigation and is documented and evaluated according to the site's Quality System.

2 BILL OF MATERIALS

Type	Component Name	Minimum Grade	Part Number(s)	Justification/Notes
Product Intermediates	INT-017	NA	NA	NA
	Payload SG3932	NA	NA	NA
Pre-packed Chromatography media	Capto MMC ImpRes	NA	TBD	NA
Buffer and solution components	Acetic acid, glacial	USP, EP	4101341 6000325	NA
	Benzyl alcohol	NF, EP, JP	4102670	NA
	Ammonium chloride	USP, EP	6004616	NA
	Ammonium Phosphate, dibasic	ACS	6004618	NA
	Ammonium phosphate, monobasic	ACS	6004617	NA
	Ethylenediamine-tetraacetic acid, disodium salt, dihydrate (EDTA)	USP	4100799 6003688	NA
	Dimethyl Sulfoxide (DMSO)	USP	4102623 6000316 6001867	NA
	L-Histidine HCl monohydrate	USP	4101011	NA
	L-Histidine	USP	4101095	NA
	Polysorbate 80	NF, EP	4100792	NA
	Sodium acetate, trihydrate, crystal	USP, EP, JP	4101342 6000326	
	N-acetyl Cysteine (NAC)	GMP Manufactured	6003593	NA
	Sodium Chloride	USP	6000324	NA

Type	Component Name	Minimum Grade	Part Number(s)	Justification/Notes
	10N Sodium hydroxide	GMP Manufactured	6002721	NA
	Sodium hydroxide, pellets	BP, Ph. Eur., JP. NF	4101097	NA
	Sodium phosphate, dibasic, hepta (7) hydrate	USP	6002094 6001964 4101092	NA
	Sodium phosphate, monobasic, monohydrate	BP, USP	4101091 6002047 6002096	NA
	Sucrose (low endotoxin)	USP, BP, JP	4102624	NA
	Tris base (tromethamine/trometamol)	USP, EP	4101399	NA
	Tris Hydrochloride	NA	4100924	NA
	Tris (2-carboxyethyl) phosphine, Hydrochloride (TCEP)	NA	6004537 6004538	NA
Filters	Pellicon 3 Ultracel capsules C Screen 30kDa MWCO	NA	6004003 6004004	0.1 m^2 0.5 m^2 1.1 m^2
	Durapore Opticap 0.2 µm	NA	NA	Sterile Grade 0.2 µm
Miscellaneous	HDPE Bottles	NA	4102052 4102055	1 L 4 L

3 IN-PROCESS HOLD TIMES

Hold times begin when product is collected and end when the next step is initiated.

Unless otherwise noted, room temperature hold times are additive to refrigerated temperature hold times but not vice versa. For example, a product that was held for 2 days at room temperature prior to being stored 1 day refrigerated would have an additive hold time of 3 days at refrigerated temperature but only 2 days at room temperature.

Table 3.1 Recommended Process Intermediate Hold Times

Process Intermediate	Room Temperature Hold Time NOR-Max (days, 15-25 °C)	Refrigerated Temperature Hold Time NOR-Max (days, 2-8 °C)	Justification/Notes
mAb Intermediate	≤ 3 days	≤ 7 days	NA
Diluted mAb Intermediate	≤ 24 hours	≤ 24 hours	NA
Quenched product	≤ 18 hours	≤ 18 hours	Product must be protected from light.
Capto MMC ImpRes Product	≤ 24 hours	≤ 24 hours	Product must be protected from light.
Pre-formulated Intermediate	≤ 24 hours	≤ 24 hours	Product must be protected from light.
Formulated Bulk	≤ 24 hours	≤ 24 hours	Product must be protected from light.

4 LOAD AND PRODUCT FILTRATION

Table 4.1 Filtration Parameters

Unit Operation	Position	Membrane Type	In-line	Volumetric Load NOR-Max (L/m^2)	Justification/ Notes
Diluted mAb Intermediate	Product Filter	Express SHC, 0.5/0.2 µm	Off-line	35	Volumetric Load NOR Max based on achieved challenge of ~35L/m^2.
Capto MMC ImpRes Chromatography (LF → Unit Op → PF)	Load Filter	Express SHC, 0.5/0.2 µm	All Steps	205	Volumetric Load NOR Max based on achieved challenge of ~203L/m^2.
	Product Filter	Express SHC, 0.5/0.2 µm	Collection	TBD	Expected challenge ~420L/m^2.
Concentration and Diafiltration (Unit Op → PF)	Product Filter	Express SHC, 0.5/0.2 µm	Off-line	TBD	Expected challenge ~115L/m^2.
Conjugated Drug Substance Formulation and Bulk Filtration (Unit Op → Bulk filter)	Bulk Filtration	Durapore Opticap, 0.2 µm	Off-line	80	Volumetric Load NOR Max based on achieved challenge of ~78L/m^2.

Millipore SHC filters may be interchanged with Pall Fluorodyne EX EDF filters of the same size. This assumes the additional filter area of the Pall filter compensates for any difference in maximum volumetric load

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5 MAB PREPARATION

Table 5.1 mAb Preparation: Process Parameters

Step	Parameter (units)	Target	AL		NOR		Justification/Notes
			Min	Max	Min	Max	
Adjustment to 10.1g/L protein concentration *20 mM sodium phosphate, 141mM sodium chloride, 1mM EDTA, pH 7.4*	Conc, (g/L)	10.1 g/L	NA	NA	9.0	11.0	~3.331-4.286 kgs of buffer to kg mAb intermediate DS solution ~ 8.7g/L protein concentration will be achieved after addition of reduction, conjugation, and quench solutions. The starting amount of mAb intermediate should be adjusted based on the amount of payload weighed. Density of diluted mAb intermediate = 1.0133 g/mL

6 MAB REDUCTION

Table 6.1 mAb Reduction: Process Parameters

Step	Parameter (units)	Target	AL		NOR		Justification/Notes
			Min	Max	Min	Max	
Temperature Control	°C	37	35	39	34	40	NA
Reduction *50mM Neutralized TCEP*	Mole TCEP to mole mAb intermediate	7.0	NA	NA	6.7	7.3	~0.0098 kg TCEP solution to kg diluted mAb intermediate solution Assume a density of 1.01 g/mL for TCEP solution.
	Time (mins)	90	85	95	80	100	Gentle mixing without foaming and entrainment of air. Incubation time ends when the volume addition of the next step is fully completed

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7 MAB CONJUGATION

Table 7.1 mAb Conjugation: Process Parameters

Step	Parameter (units)	Target	AL		NOR		Justification/Notes
			Min	Max	Min	Max	
Temperature Control during payload addition and incubation	°C	21.0	19.0	37.0	18.0	39.0	DMSO and payload can be added before target temp reached Exothermic reaction. Ensure temperature control is adequate to avoid exceeding NOR max. Control rate of payload addition to help mitigate temperature increase.
Conjugation *25mM payload in DMSO*	Mole payload to mole mAb intermediate	11.0	NA	NA	10.5	11.5	Target 11x molar ratio. ~0.0343 kg payload per kg **diluted mAb intermediate** solution Payload is dissolved in DMSO solution when delivered to the reduced mAb intermediate. Assume a density of 1.129 g/mL for payload stock solution.
	DMSO (vol) / Total solution (vol)	0.10	NA	NA	0.09	0.11	Target is 10% vol including payload (*i.e.* DMSO added + payload added = 10% of total volume) ~0.0884 kg DMSO per kg **diluted mAb intermediate** solution Assume a density of 1.10 g/mL for DMSO
	Incubation time (mins)	60	NA	NA	50	70	Gentle mixing without foaming and entrainment of air. Incubation time ends when the volume addition of the next step is fully completed.

Table 7.1 mAb Conjugation: Process Parameters

Step	Parameter (units)	Target	AL		NOR		Justification/Notes
			Min	Max	Min	Max	
Quenching *100mM NAC*	Mole NAC to mole payload added	5.0	NA	NA	4.75	5.25	Target 5x molar ratio of NAC to payload ~0.0323 kg NAC solution per kg **crude conjugate** Assume density of 1.00 mg/mL for NAC stock solution
	Time (mins)	15	NA	NA	0	1080	Quench time ends when full volume is loaded onto Capto MMC ImpRes column. Target to complete load by the target time. Gentle mixing without foaming and entrainment of air
	Temperature Control during incubation (°C)	20.0	NA	NA	18.0	25.0	Column load and binding can be impacted if temperature is too high.

NOTE: KEEP LOAD PRODUCT AND COLLECTION VESSEL COVERED FROM LIGHT AS MUCH AS POSSIBLE DURING THE PROCESS. COVER SAMPLES IMMEDIATELY AFTER ALIQUOTING.

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8 CAPTO MMC IMPRES CHROMATOGRAPHY

Table 8.1 Capto MMC ImpRes Chromatography: Process Parameters

Step	Parameter (units)	Target	AL		NOR		Justification/Notes
			Min	Max	Min	Max	
General	Media	Capto MMC ImpRes					Prep per appropriate SOP
	Bed height (cm)	20	NA	NA	18	22	NA
Equilibration 1 *50 mM ammonium phosphate, pH 6.0*	Linear velocity (cm/hr)	200	190	210	NA	220	AL is ± 5% of set-point
	Volume (CV)	5.0	2.9	6.0	2.5	NA	NA
	pH, effluent, (endpoint check) [a]	NA	NA	NA	5.9	6.1	NA
	Conductivity, effluent, (endpoint check) (mS/cm) [a]	NA	NA	NA	NA	NA	Run and record
Load *Quenched Conjugated Product*	Linear velocity (cm/hr)	150	142	158	NA	220	AL is ± 5% of set-point
	Protein load (g / L resin)	NA	NA	NA	10	15	Max is based on binding capacity. AL-Min is based on productivity. Utilize the maximum protein load on the column per cycle when possible.
Equilibration 2 *50 mM ammonium phosphate, pH 6.0*	Linear velocity (cm/hr)	200	190	210	NA	220	AL is ± 5% of set-point
	Volume (CV)	5.0	4.9	NA	4.5	NA	NA
Elution	Linear velocity (cm/hr)	200	190	210	NA	220	AL is ± 5% of set-point

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Table 8.1 Capto MMC ImpRes Chromatography: Process Parameters

Step	Parameter (units)	Target	AL		NOR		Justification/Notes
			Min	Max	Min	Max	
Buffer A: *50 mM ammonium phosphate, pH 6.0* Buffer B: 51 *mM ammonium phosphate, 1020 mM ammonium chloride, pH 6.0*	Volume (CV)	10	NA	NA	NA	NA	Start with 100% A and go to 100% B over 10CV length. Proceed to strip as soon as collection is complete.
	Buffer B %, initial	0	NA	NA	NA	NA	NA
	Buffer B %, final	100	NA	NA	NA	NA	NA
	Post Gradient hold (CV)	NA	NA	NA	NA	5.0	100% buffer B Proceed to Post Gradient hold if collection is still ongoing at the conclusion of the gradient. Proceed to strip as soon as collection is complete.
	Start collection OD_{280}	0.5	0.5	1.0	0.5	NA	Volume before peak ~ 5 CV
	End collection OD_{280}	0.5	0.5	1.0	0.5	NA	Product volume = ~ 6.5 CV
Strip *50 mM Tris, 500 mM sodium chloride, pH 7.3*	Linear velocity (cm/hr)	200	190	210	NA	220	AL is ± 5% of set-point
	Volume (CV)	3.0	2.9	4.0	2.5	NA	NA
Caustic Strip *0.5 N sodium hydroxide*	Linear velocity (cm/hr)	200	190	210	NA	220	AL is ± 5% of set-point
	Volume (CV)	3.0	2.9	4.0	2.5	NA	Sanitization buffer
	Conductivity, effluent, (endpoint check) (mS/cm)[a]	NA	NA	NA	84	113	NA
End of Cycle							

Table 8.1 Capto MMC ImpRes Chromatography: Process Parameters

Step	Parameter (units)	Target	AL		NOR		Justification/Notes
			Min	Max	Min	Max	
Sanitization Hold	Contact time (minutes)	35	30	60	15	120	NA
Storage *2% (v/v) benzyl alcohol, 100 mM sodium acetate, pH 5.0*	Linear velocity (cm/hr)	200	190	210	NA	220	AL is ± 5% of set-point
	Volume (CV)	3.0	2.9	4.0	2.5	NA	NA
	pH, effluent, (endpoint check)[a]	NA	NA	NA	4.8	5.2	NA

[a] Output Parameter

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9 CONCENTRATION AND DIAFILTRATION

Table 9.1 Concentration and Diafiltration: Process Parameters

Step	Parameter (units)	Target	AL		NOR		Justification/Notes
			Min	Max	Min	Max	
General	Media	Pellicon 3 Ultracel Capsules C Screen 30kDa MWCO					Prep per appropriate SOP
	Feed pressure (psig)	NA	NA	55	NA	60	NA
Equilibration *Diafiltration buffer:* *20 mM Histidine/histidine-HCl,* *pH 6.0*	pH [a]	6.0	5.8	6.2	5.7	6.3	NA
Concentration 1 Control Method: TMP and Feed Flow Control	Membrane protein load (g/m^2)	NA	NA	100	NA	225	(Mass of protein)/(area of ultrafiltration membrane)
	TMP (psig)	20	NA	30	NA	35	NA
	Feed flow ($L/min/m^2$)	5.0	4.0	6.0	3.0	7.0	NA
	Conc. 1, final (g/L)	20	15	25	15	30	May be directly measured or calculated from permeate or retentate volume
Diafiltration *Diafiltration buffer:*	TMP (psig)	20	NA	30	NA	35	NA
	Feed flow ($L/min/m^2$)	5.0	4.0	6.0	3.0	7.0	NA

Table 9.1 Concentration and Diafiltration: Process Parameters

Step	Parameter (units)	Target	AL		NOR		Justification/Notes
			Min	Max	Min	Max	
20 mM Histidine/histidine-HCl, pH 6.0 Control Method: TMP and Feed Flow Control	Diafiltration volumes	6.0	6.0	7.0	6.0	NA	(kg buffer)/(kg product)
	Permeate pH, final [a]	6.0	NA	NA	5.8	6.2	NA
	Retentate pH, final [a]	6.0	NA	NA	5.8	6.2	NA
Concentration 2 Control Method: TMP and Feed Flow Control	TMP (psig)	20	NA	30	NA	35	NA
	Feed flow (L/min/m^2)	NA	NA	6.0	NA	7.0	NA
	Conc. 2, final (g/L)	35	27	39	25.4	40	May be directly measured or calculated from permeate or retentate volume
Recirculation	Feed pressure (psig)	20	NA	55	NA	60	NA
	Feed flow (L/min/m^2)	NA	NA	6.0	NA	7.0	NA
	Recirculation time (min)	5	NA	30	NA	120	NA
Recovery Flush *Diafiltration buffer:*	Feed pressure (psi)	20	18	22	NA	30	NA
	Feed flow (L/min/m^2)	NA	NA	6.0	NA	7.0	NA

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Table 9.1 Concentration and Diafiltration: Process Parameters

Step	Parameter (units)	Target	AL		NOR		Justification/Notes
			Min	Max	Min	Max	
20 mM Histidine/histidine-HCl, pH 6.0 Method: Buffer flush of system hold up volume	Recovery conc. (g/L)	30	NA	NA	25.4	40	NA
Dilution to Pre-formulated intermediate *Diafiltration buffer: 20 mM Histidine/histidine-HCl, pH 6.0*	PFI conc. (g/L)	26.7	26	27.4	25.4	28	NA
	pH [a]	6.0	5.9	6.1	5.8	6.2	NA

[a] Output parameter.

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10 FORMULATION AND BULK FILTRATION

Table 10.1 Formulation and Bulk Filtration: Process Parameters

Step	Parameter (units)	Target	AL		NOR		Justification/Notes
			Min	Max	Min	Max	
EAB/ FSS spike (of PFI to FB): *EAB: 20 mM histidine, 957 mM sucrose, 0.08% (w/v) PS-80, pH 6.0* (density 1.124 g/mL)	Ratio of EAB mass to PFI mass (kg/kg)	0.376	0.372	0.380	0.367	0.385	FB/DS: *20.0 g/L AZD9592 in 20mM Histidine, 240 mM sucrose, 0.02% PS 80 (w/v), pH 6.0* (density 1.033 g/mL)
	Protein concentration, endpoint check (g/L)	20.0	18.0	22.0	NA	NA	NA
	Osmolality (mOsm/kg)[a]	300	NA	NA	255	345	NA
0.2µm Bulk Filtration of FB to DS	Capacity (L/m^2)	NA	NA	NA	NA	80	Challenge achieved at SAFC 78L/m^2.
	Drug substance container	HDPE bottle					DS filled HDPE bottles need to be placed in bag to be protected from light.
	Fill volume per container (%)	70	NA	NA	NA	90	If multiple containers are required, target equivalent volume in each.
	Drug substance storage temperature (°C)	-80	NA	NA	NA	-70	Drug substance can be held 3 days at 2-8°C in mylar bags prior to freezing.
Integrity Test *0.2 µm Filter*	Integrity Test Result (PASS/FAIL)[a]	PASS	NA	NA	PASS	PASS	Perform per appropriate SOP. Integrity test is an in-process control that must pass for the lot to be released

[a] Output parameter

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11 BUFFERS AND SOLUTIONS

Buffer and solution stability studies are a commercial process definition activity. However, buffers and solutions with completed studies because of prior experience or a stability concern have stated hold times below. The studies were performed by holding buffers and solutions at ambient temperature in stainless steel containers. The studies were designed to test chemical stability for longer than the expected maximum time for routine storage. These times may be extended by an appropriate study unless otherwise noted. The studies do not address microbial control, which should be defined by the facility. Assays used to measure stability included pH and conductivity. In some cases, the hold time is measured directly (direct). In other cases, the hold time is supported by the hold time of similar buffers if the proposed buffer falls within our current buffer matrix (matrix). Please see DEV000_GB_0533 for reference.

Table 11.1 Buffer and Solution References

Buffer Name	Recipe Table	Doc Num/ AZ PN	Unit Operation	Steps
20 mM sodium phosphate, 141 mM sodium chloride, 1 mM EDTA, pH 7.4	Table 11.3	8009867	mAb Preparation	mAb preparation diliution
50 mM Neutralized TCEP	Table 11.2	AZDoc0213726/ 8009045	mAb Reduction	Reduction
25 mM SG3932 payload in DMSO	Table 11.4	N/A	mAb Conjugation	Conjugation
100 mM NAC	Table 11.5	AZDoc0213727/ 8009047	mAb Conjugation	Quenching
50 mM ammonium phosphate, pH 6.0	Table 11.6	8009868	Capto MMC ImpRes	Equilibration 1
				Equilibration 2

				Elution Buffer A
51 mM ammonium phosphate, 1020 mM ammonium chloride, pH 6.0	Table 11.7	8009869	Capto MMC ImpRes	Elution Buffer B
50 mM Tris, 500 mM sodium chloride, pH 7.3	Table 11.8	8007146	Capto MMC ImpRes	Strip
0.5 N sodium hydroxide	Table 11.11	A0013-D/ A0013-D	Capto MMC ImpRes	Caustic Strip
2% (v/v) benzyl alcohol, 100 mM sodium acetate, pH 5.0	Table 11.12	8002256/ A-005212	Capto MMC ImpRes	Storage
20 mM Histidine/histidine-HCl, pH 6.0	Table 11.10	A-0643	Concentration and Diafiltration	Equilibration
				Diafiltration
				Recovery Flush
				Dilution
20 mM histidine/histidine-HCl, 957 mM sucrose, 0.08% (w/v) PS-80, pH 6.0	Table 11.9	8010085	Formulation and Bulk Filtration	EAB/FSS spike

Table 11.2 50 mM TCEP

MEDI PN	8009045		
Density Corrected	Yes		
Specification	**Target**	**NOR**	
pH	7.0	6.5 – 7.5	
Conductivity (mS/cm at 25°C)	TBD (run and record)	TBD	
Osmolality (mOsmol/kg)	TBD (run and record)	TBD	
Density (g/mL)	1.01	NA	
Chemical	**MEDI PN**	Amount by Volume (mL/kg)	Amount by Weight (g/kg)
TCEP.HCl	6004537 6004538	NA	14.19
WFI	NA	969.1	969.1
10N sodium hydroxide**	6002721	15.0	16.7
Total		1000 g/kg	

*Prepare and use within 24 hours

**Mass of hydroxide required may vary from target, requiring titration

Table 11.3 20 mM sodium phosphate, 141 mM sodium chloride, 1mM EDTA, pH 7.4

MEDI PN	8009867/B-811		
Density Corrected	Yes		
Specification	**Target**	**NOR**	
pH	7.4	7.3 - 7.5	
Conductivity (mS/cm at 25°C)	TBD (run and record)	TBD	
Osmolality (mOsmol/kg)	TBD (run and record)	TBD	
Density (g/mL)	1.007	NA	
Chemical	**MEDI PN**	**Amount by Volume (mL/kg)**	**Amount by Weight (g/kg)**
Sodium Phosphate, Dibasic, Heptahydrate	6002094 6001964 4101092	NA	4.77
Sodium Phosphate, Monobasic, Monohydrate	4101091 6002047 6002096	NA	0.30
Sodium Chloride	6000324	NA	8.18
EDTA disodium salt dihydrate	4100799 6003688	NA	0.37
WFI	NA	NA	986.38
Total		1000 g	

Table 11.4 25 mM payload SG3932 in DMSO

MEDI PN	N/A		
Density Corrected	Yes		
Specifications	**Target**	**NOR**	
pH	NA	TBD	
Conductivity (mS/cm at 25°C)	NA	TBD	
Osmolality (mOsmol/kg)	NA	TBD	
Density (g/mL)	1.129	NA	
Chemical	**MEDI PN**	**Amount by Volume (mL/kg)**	**Amount by Weight (g/kg)**
Payload SG3932	NA	NA	25.43
DMSO	4102623 6000316 6001867	NA	974.57
Total		1000 g/kg	

[a] Payload purity should be accounted for the solution preparation.
*Prepare and use within 24 hours
**Solution should be covered to protect from light

Table 11.5 100 mM NAC

MEDI PN:	8009047		
Density Corrected	Yes		
Specification	**Target**	**NOR**	
pH	NA	NA	
Conductivity (mS/cm at 25°C)	TBD (run and record)	TBD	
Density (g/mL)	1.00	NA	
Chemical	**MEDI PN**	**Amount by Volume (mL/kg)**	**Amount by Weight (g/kg)**
NAC	6003593	NA	16.32
WFI	NA	983.68	983.68
Total		1000 g/kg	

*Prepare and use within 24 hours

Table 11.6 50 mM ammonium phosphate pH 6.0

MEDI PN:	8009868/B-812		
Density Corrected	Yes		
Specification	**Target**	**NOR**	
pH	6.0	5.9 - 6.1	
Conductivity (mS/cm at 25°C)	TBD (run and record)	TBD	
Density (g/mL)	1.002	NA	
Chemical	**MEDI PN**	**Amount by Volume (mL/kg)**	**Amount by Weight (g/kg)**
Ammonium phosphate, dibasic	6004618	NA	0.85
Ammonium phosphate, monobasic	6004617	NA	5.01
WFI	NA	NA	994.14
Total		1000 g/kg	

Table 11.7 51 mM ammonium phosphate, 1020 mM ammonium chloride, pH 6.0

MEDI PN:	8009869/B-813		
Density Corrected	Yes		
Specification	**Target**	**NOR**	
pH	6.0	5.9 - 6.1	
Conductivity (mS/cm at 25°C)	TBD (run and record)	TBD	
Density (g/mL)	1.018	NA	
Chemical	**MEDI PN**	**Amount by Volume (mL/kg)**	**Amount by Weight (g/kg)**
Ammonium phosphate, dibasic	6004618	NA	2.89
Ammonium phosphate, monobasic	6004617	NA	3.24
Ammonium chloride	6004616	NA	53.49
WFI	NA	NA	940.38
Total		1000 g/kg	

Table 11.8 50 mM Tris, 500 mM sodium chloride, pH 7.3

MEDI PN:	8007146/ B-663		
Density Corrected	Yes		
Specification	**Target**	**NOR**	
pH	7.3	7.1 – 7.5	
Conductivity (mS/cm at 25°C)	NA	43 - 56	
Density (g/mL)	1.021	NA	
Chemical	**MEDI PN**	**Amount by Volume (mL/kg)**	**Amount by Weight (g/kg)**
Tris base	4101399	NA	0.65
Tris HCl	4100924	NA	6.87
Sodium Chloride	6000324	NA	28.62
WFI	NA	NA	963.9
Total		1000 g/kg	

Table 11.9 20 mM histidine, 957 mM sucrose, 0.08% (w/v) polysorbate 80, pH 6.0

MEDI PN:	8010085/B-532		
Density Corrected	Yes		
Specification	**Target**	**NOR**	
pH	6.0	5.8 – 6.2	
Conductivity (mS/cm at 25°C)	TBD (run and record)	TBD	
Density (g/mL)	1.124	NA	
Chemical	**MEDI PN**	**Amount by Volume (mL/kg)**	**Amount by Weight (g/kg)**
L-histidine	4101095	NA	1.4
L-histidine HCl monohydrate	4101011	NA	1.82
Sucrose	4102624	NA	291.58
10% (w/v) PS-80 stock	A-0647	7.03	7.1
WFI	NA	NA	698.1
Total		1000 g/kg	

Table 11.10 20 mM Histidine/Histidine-HCl, pH 6.0

MEDI PN/ Document No.	A-0643		
Density Corrected	Yes		
Specification	**Target**	**Range**	
pH	6.0	5.8- 6.2	
Density (g/mL)	1.00	NA	
Chemical	**MEDI PN**	**Amount (mL/kg)**	**Amount (g/kg)**
L-histidine-HCl monohydrate	4101011	NA	2.21
L-histidine	4101095	NA	1.47
WFI	NA	NA	996.32
Total		1000 g/kg	

Table 11.11 0.5 N sodium hydroxide

MEDI PN/ Document No.	A0013-D/A0013-D		
Density Corrected	Yes		
Specification	**Target**	**Range**	
pH	NA	NA	
Conductivity (mS/cm at 25°C)	NA	84-113	
Osmolality (mOsm/kg)	NA	NA	
Density (g/mL)	1.02	NA	
Chemical	**MEDI PN**	**Amount (mL/kg)**	**Amount (g/kg)**
Sodium hydroxide, pellets	4101097	NA	19.61
WFI	NA	NA	980
Total		1000 g/kg	

Table 11.12 2% (v/v) benzyl alcohol, 100 mM sodium acetate, pH 5.0

MEDI PN/ Document No.	8002256/A-005212		
Density Corrected	Yes		
Specification	**Target**	**Range**	
pH	5.0	4.8- 5.2	
Conductivity (mS/cm at 25°C)	NA	4.6-5.6	
Osmolality (mOsm/kg)	NA	NA	
Density (g/mL)	1.00	NA	
Chemical	**MEDI PN**	**Amount (mL/kg)**	**Amount (g/kg)**
Sodium acetate, trihydrate	4101342	NA	9.31
Acetic acid, glacial	6000325	1.81	1.90
Benzyl alcohol	4102670	20.0	20.88
WFI	NA	NA	968
Total		1000 g/kg	

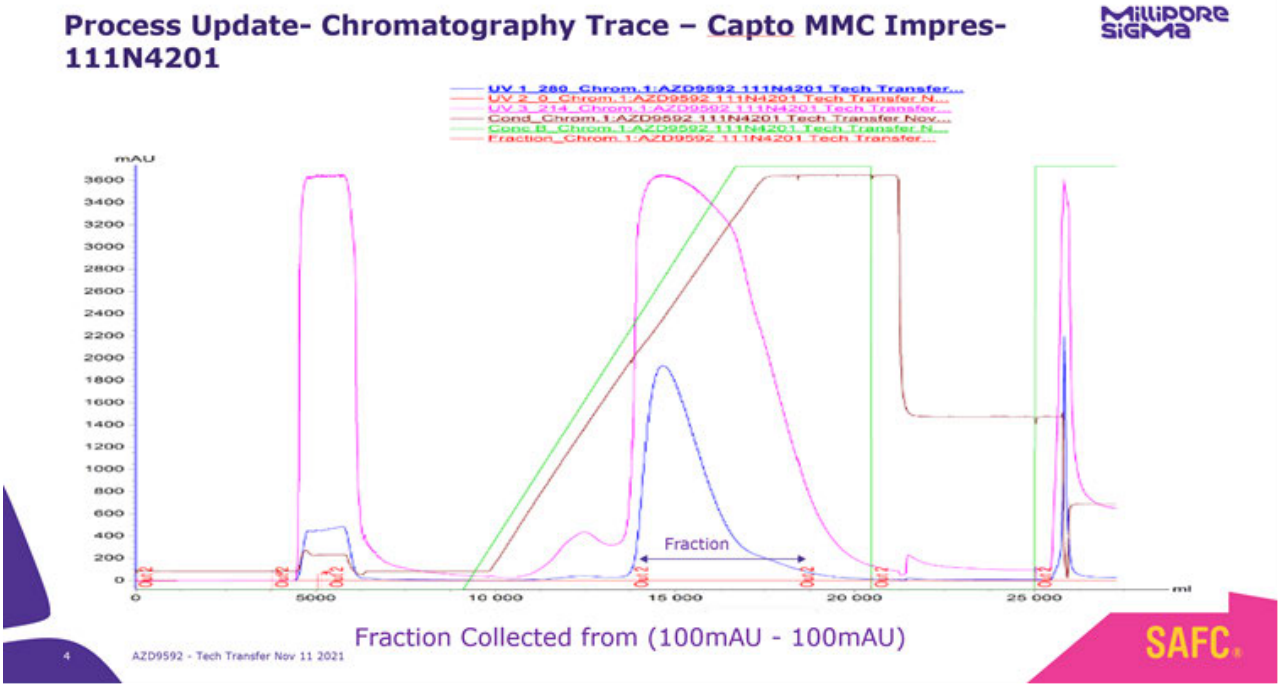
12 REFERENCE CHROMATOGRAMS

Sample chromatograms representing the chromatography step used in the purification process are presented below.

Table 12.1 CaptoMMC ImpRes Chromatogram

Resin:	CaptoMMC ImpRes
Column Dimensions:	7 cm (D) x 19.8 cm (H) – 761.99 mL (CV)
Notes:	

Figure 12-1 Capto MMC ImpRes Chromatogram



13 PROCESS CHANGES

Table 13.1 lists changes in the AZD9592 Conjugation Process Description from edition 001 to edition 002. These changes do not warrant a process change from Process 1.

Unit Operation	Step	Parameter	Original	New	Summary/Justification
mAb conjugation	Temperature Control	ºC	**ED001:** Target = 20.0 AL = NA	**ED002:** Target = 21.0 AL = 19 – 37	Increased target to robustly stay within range and added AL ranges for manufacturing operations.

14 VERSION HISTORY

Version	Reason
1.0	New document
2.0	• Table 3.1: Updated hold time NOR for quenched product at 2-8 and 15-25C to <=1 hour from <=2 hours. • Revised temperature set point during payload addition and incubation and added AL in Table 7.1. • Corrected PN of sodium phosphate, dibasic, heptahydrate in Table 11.3 and BOM. • Corrected PN of 50 mM Tris, 500 mM sodium chloride, pH 7.3 in Table 11.8. • Added PN of 10% PS80 stock solution and added volumetric addition ratio in Table 11.9. • Added buffer tables for 0.5 N sodium hydroxide (Table 11.11) and 2% (v/v) benzyl alcohol, 100 mM sodium acetate, pH 5.0 (Table 11.12). • Updated Conductivity spec to TBD for 8009867/B-811 (Table 11.3), 8009868/B-812 (Table 11.6), and 8009869/B-813 (Table 11.7). • Table 1.1: Added units for extinction coefficient. • Updated decimal spaces for NOR Conc (g/L) in Table 5.1, and for Temperature Control in table 7.1. • Changed “Run and Record” to NA in table 8.1. • Table 8.1 & 11.1: Updated name of storage buffer to align with DEV000_GB_0469 ED 009 • Added “Acetic acid, glacial”, “Benzyl alcohol” and “Sodium acetate, trihydrate, crystal” to BOM because they are components of “2% (v/v) benzyl alcohol, 100 mM sodium acetate, pH 5.0”.
3.0	• Updated Equilibration 1 CV target from 3 to 5CVs and AL max from 4.0 to 6.0 CVs to ensure column is fully equilibrated. • Updated Table 11.1 to use PN 8007146 instead of PN 8009870 in order to align with currently available PN. • Updated Osmo Min and Max Specs from AL to NOR as the osmolality is not included in the master spec.
4.0	• Table 3.1: Updated Process Intermediate Hold Times for Quenched product, Capto MMC Impres Product, and Pre-formulated Intermediate

Document Approvals

Business Approval	[redacted]
Business Approval	[redacted]